Architecture, resolution and calibration in pediatric pneumonia image classification

An exploratory reproducible benchmark

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Abstract

Purpose. To compare conventional pretrained image classifiers under a fixed development protocol, separating ranking performance from calibration and threshold-dependent classification. This report concerns dataset labels on pediatric chest radiographs and does not establish clinical diagnosis or utility.

Methods. A decoded-image audit and conservative filename grouping retain 3,815 training, 456 model-selection, 482 calibration and 618 test images. EfficientNet-B0, DenseNet121 and ResNet18 are trained with three seeds each at 224 and 320 pixels (18 trials). Mean selection AUROC chooses the architecture; its seed probabilities are averaged. A positive temperature and a Youden threshold are fit on the separate calibration partition before test evaluation. All comparisons on the original test set are exploratory because prior results were inspected.

Results. All 18 trials completed. Development-only selection across both resolutions chose the 320-pixel EfficientNet-B0 ensemble. Its test AUROC was 0.9640 (95% group-bootstrap interval 0.9470 to 0.9787), accuracy 87.38%, sensitivity 99.74% and specificity 66.67%. The paired AUROC difference versus the 224-pixel ResNet18 ensemble was 0.0059 (95% interval -0.0055 to 0.0170), which does not establish improvement. CPU test reproduction changed zero classifications for either ensemble.

Interpretation. The study is an exploratory benchmark with explicit audit and reproducibility checks. External testing, independently verified patient identities and clinical utility remain unestablished. A longer training duration alone does not resolve these limitations.

Introduction

Pediatric pneumonia image classification is frequently studied using the Guangzhou chest-radiograph resource distributed by Kermany and colleagues [1]. Performance on one source dataset does not demonstrate transportability. Zech and colleagues reported variable pneumonia-model generalization across datasets [2], and a recent pediatric preprint explicitly examined discrimination, calibration and fixed-threshold transport across countries [3]. These findings motivate reporting several complementary outcomes rather than treating accuracy as a complete measure of model quality.

This project asks how architecture, input resolution and random initialization affect discrimination and the behavior of a development-selected decision rule under a constrained compute budget. It compares three established architectures, not a newly invented network. Temperature scaling is an established calibration technique [4]. The intended contribution is a transparent, reproducible benchmark and an honest account of its limitations; novelty sufficient for a particular publication venue has not been established.

Materials and methods

Computing environment. Both resolution arms ran sequentially on one Tesla T4, with PyTorch 2.11.0+cu128, torchvision 0.26.0+cu128, NumPy 2.1.3, pandas 2.3.3 and scikit-learn 1.6.1. The local CPU

development environment differs; its dependency snapshot is recorded separately. GPU and CPU agreement is evaluated numerically rather than assumed to be bitwise identical.

Study design. This is a retrospective computational study of public research images. No patient care is involved. The preliminary baseline and follow-up were completed before this benchmark was designed. Their original test outcomes were already known; therefore the current source test evaluation cannot be described as an untouched confirmatory test.

Data audit. The attached Kaggle resource contains 5,856 decoded images. The audit hashes grayscale dimensions and pixels, detects conflicting labels for identical images, removes exact duplicate test images, and excludes development images sharing a derived group or pixel hash with test. Development duplicates are removed deterministically. A total of 5,371 images remain. Near-duplicate detection and clinical patient-ID verification are not established by this exact-hash check.

Grouping limitation. Groups are derived from filename patterns IM, NORMAL2-IM and person-number. The person-number rule groups bacterial and viral filenames together, although numbers may be reused across disease namespaces. This conservative grouping may merge unrelated children and exclude additional development images. The groups should not be called verified patient identities. The frozen split is retained to preserve the planned comparison.

Partition	Images	Derived groups	Normal	Pneumonia
Training	3815	2153	1078	2737
Model selection	456	270	137	319
Calibration	482	268	133	349
Source test	618	426	231	387

The earlier 938-image validation cohort is divided by group and class into selection and calibration partitions using a fixed hash ordering. Explicit original held-out group membership and SHA-256 cohort digests prevent scikit-learn version changes from silently changing the training or test images. Every pair of partitions must have zero exact-pixel-hash and derived-group overlap.

Image processing. Images are converted to RGB, resized with bicubic interpolation and padded to 224 by 224 or 320 by 320 pixels while preserving the entire image and aspect ratio. Training applies random affine augmentation with at most 7 degrees of rotation, translation fractions 0.03 and scale 0.95 to 1.05. Tensors use ImageNet channel means and standard deviations. Lossless PNG caching of resized images avoids repeated original-image decoding. Local checks confirm equality of cached and direct preprocessing tensors.

Model training. Torchvision ImageNet V1 weights [7] initialize EfficientNet-B0, DenseNet121 and ResNet18; each receives one binary output. Seeds are 42, 123 and 2026, including each trial's data-loader order and augmentation seeds. AdamW uses a backbone learning rate of 0.0001, head learning rate of 0.001 and weight decay of 0.0001. Weighted binary cross entropy assigns the positive class the training normal-to-pneumonia count ratio. Batch size is 32. Training uses mixed precision, whereas evaluation uses float32.

The scheduler halves the learning rate after a plateau of two epochs. Selection AUROC determines checkpoint replacement with a minimum improvement of 0.00001; training stops after eight unimproved epochs or a time limit, with a maximum of 60 epochs per trial. Architecture order rotates across seeds. The

270-minute audit-and-training budget is allocated dynamically over remaining trials. This is a shared time-budget comparison, not a guarantee of equal epochs or identical FLOPs. Partial final epochs and stop reasons are retained in the histories. One GPU is used even if Kaggle allocates a T4 pair.

Selection and calibration. Architecture is selected by mean model-selection AUROC over all three seeds; exact ties use alphabetical order. Seed sigmoid probabilities are averaged. The averaged probability is clipped to [0.0000001, 0.9999999], converted to a logit and divided by a positive scalar temperature. Temperature minimizes calibration-set negative log likelihood, with log-temperature bounded between -3 and 3. Maximum Youden J on calibration scores determines the binary cutoff. The inference configuration is saved before test evaluation. No test-based choice of model, temperature or threshold is permitted.

Outcome measures. Primary descriptive outcomes are AUROC and sensitivity and specificity at the frozen development threshold. Average precision, accuracy, Brier score, log loss and confusion counts supplement them. Fixed raw-score threshold 0.5 results are also retained. Positive temperature scaling preserves ordering apart from clipping or numerical ties; an AUROC gain is not an intended calibration effect. Calibration and threshold fitting share one calibration partition, so in-sample calibration performance should not be interpreted as unbiased future performance.

Uncertainty and verification. Paired comparisons retain image-hash alignment and resample filename-derived groups with replacement. Percentile intervals describe source test sampling conditional on the fitted models; they omit training and model-selection uncertainty. Three seeds provide limited descriptive evidence of training variation. No multiplicity-adjusted confirmatory claims are planned. Artifact checks reconstruct the selected ensemble from member scores, recompute metrics, and verify split identity and exported checkpoint paths.

Resolution extension. Before the new benchmark test results were observed, an exploratory 320-pixel arm was specified with the same nine architecture-seed combinations and frozen cohorts. It runs sequentially after the 224-pixel benchmark, with a 180-minute cap within the combined compute budget. Each arm selects its architecture independently from development data. A combined choice, if reported, uses mean selection AUROC only, breaking ties by lower resolution then alphabetical architecture. All arms remain reportable; expanded selection can increase optimism. Both arms completed, with Kaggle job runtimes of 3,048.6 and 5,026.1 seconds (134.58 minutes combined, including job overhead). All 18 trials stopped by the early-stopping rule rather than exhausting their caps. Training duration was not extended after convergence merely to fill the five-hour work session.

Similarity screening. A separate CPU audit compares 64-bit low-frequency DCT perceptual hashes across partitions, with the DC bit zero and a Hamming-distance cutoff of six. This is a candidate screen, not validated duplicate detection. It is not used to alter the frozen experiment cohorts.

Results

The completed preliminary baseline and follow-up used the earlier shared validation partition for checkpoint and threshold selection. They provide project context, not a controlled estimate of the effects of the new separated calibration design. The follow-up selected a single seed-123 model rather than its averaged ensemble.

Source test metric	Baseline	Selected follow-up
AUROC	0.97061	0.96844
Average precision	0.97686	0.97306
Accuracy	80.42%	88.19%
Sensitivity	100.00%	99.22%
Specificity	47.62%	69.70%

Accuracy increased from 80.42% to 88.19% under the separately selected thresholds, but AUROC decreased from 0.97061 to 0.96844. Sensitivity decreased from 100.00% to 99.22%, while specificity increased from 47.62% to 69.70%. Brier score worsened from 0.13507 to 0.14308. At a common raw threshold of 0.5, accuracy was 84.47% and 84.30%, respectively. These observations do not support an across-the-board improvement claim.

224-pixel individual trials. The following values describe individual calibrated models, not their ensemble.

Architecture and seed	Epochs	Selection AUROC	Test AUROC	Test Brier
efficientnet_b0_seed_42	24	0.99799	0.96059	0.13766
densenet121_seed_42	14	0.99902	0.95453	0.11077
resnet18_seed_42	19	0.99842	0.95667	0.10851
densenet121_seed_123	14	0.99799	0.95446	0.11730
resnet18_seed_123	25	0.99844	0.94193	0.13623
efficientnet_b0_seed_123	18	0.99838	0.97021	0.13945
resnet18_seed_2026	15	0.99883	0.96237	0.14971
efficientnet_b0_seed_2026	23	0.99760	0.96188	0.13759
densenet121_seed_2026	26	0.99831	0.96713	0.13709

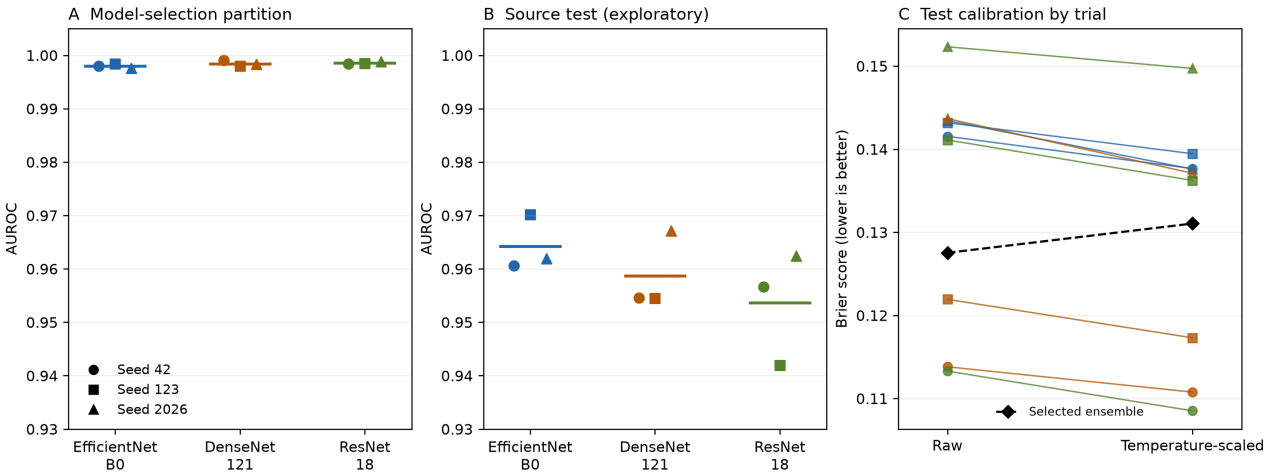


Figure 1. All nine 224-pixel trials. Colors identify architectures; symbols identify seeds. Horizontal segments in A and B show the arithmetic mean of individual-trial AUROCs, not ensemble AUROC or uncertainty intervals. C connects each trial's raw and calibrated Brier score; the black dashed line is the separately calibrated, development-selected ensemble. All test results are exploratory.

All trial-level test values are reported after development-only selection. They must not be used to retroactively replace the selected architecture. The original Kaggle outputs are authoritative. These compact results were transcribed programmatically from the fully expanded visible JSON preview, with field-count and confusion-count checks; the complete new archive has not yet been downloaded locally.

The selected ResNet18 ensemble used temperature 0.67293 and cutoff 0.72091. Its confusion counts were TN 154, FP 77, FN 1 and TP 386. Calibration worsened source-test Brier score from 0.12752 to 0.13105 and log loss from 0.97988 to 1.27754. Its test AUROC is lower than either earlier selected experiment. Thus, this comparison does not establish that more architectures, separate calibration, or additional training improved general performance.

Metric	224-pixel ensemble (95% interval)	Difference versus prior follow-up (95% paired interval)
AUROC	0.9581 [0.9395, 0.9753]	-0.0103 [-0.0258, 0.0053]
Average precision	0.9614 [0.9355, 0.9817]	-0.0116 [-0.0333, 0.0085]
Accuracy	87.38% [84.47, 90.32]	-0.81 pp [-2.56, 0.84]
Sensitivity	99.74% [99.16, 100.00]	0.52 pp [0.00, 1.30]
Specificity	66.67% [60.37, 73.02]	-3.03 pp [-7.87, 1.37]
Brier score	0.1310 [0.1031, 0.1599]	-0.0120 [-0.0242, -0.0006]

Intervals use 1,000 paired filename-group bootstrap replicates over 426 groups and 618 identical test images. Differences are the new ResNet18 ensemble minus the earlier selected EfficientNet-B0 seed-123 model. Percentage-point differences are marked pp. AUROC and accuracy intervals span zero, which neither proves equivalence nor establishes improvement. The Brier interval is negative, but the comparison is exploratory and unadjusted for multiplicity; these intervals omit training and model-selection uncertainty. Kaggle analysis version 360892651 completed successfully.

Calibration-only group resampling (1,000 replicates, 268 filename groups) yielded threshold percentiles 2.5/25/50/75/97.5 of 0.18496/0.72091/0.72091/0.73699/0.94266. The fitted temperature was held fixed, so temperature-fitting uncertainty is omitted. The deployed cutoff was unchanged.

Independent CPU reproduction passed. All nine checkpoints loaded strictly and passed a forward check. Decoded-pixel identity was verified for all 618 test images. Fresh ensemble inference changed 0 classifications; the maximum calibrated-score difference was 0.00000441, below the prespecified 0.0001 tolerance. This establishes engineering reproduction of the saved model on the same source cohort, not independent clinical or external validation. Kaggle verification version 360888922 completed in 386.9 seconds.

Resolution comparison

320-pixel individual trials. All nine planned trials completed. The same architecture-seed combinations and image identities were retained across resolutions. These are matched protocol comparisons, with adaptive early stopping; they do not isolate resolution from differences in selected checkpoints or training duration.

Architecture and seed	Epochs	Selection AUROC	Test AUROC	Test Brier
efficientnet_b0_seed_42	20	0.99902	0.96147	0.14331
densenet121_seed_42	21	0.99767	0.97414	0.10280
resnet18_seed_42	20	0.99789	0.95754	0.13065
densenet121_seed_123	11	0.99698	0.96691	0.19557
resnet18_seed_123	26	0.99883	0.96507	0.13078
efficientnet_b0_seed_123	22	0.99908	0.96044	0.13163
resnet18_seed_2026	13	0.99812	0.96027	0.14898
efficientnet_b0_seed_2026	26	0.99915	0.95880	0.15209
densenet121_seed_2026	15	0.99860	0.97450	0.13072

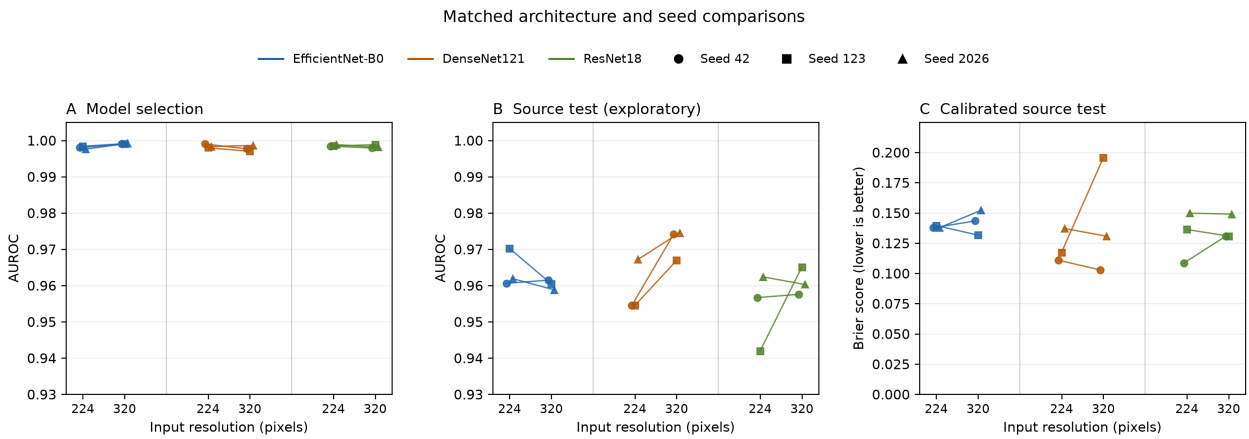


Figure 2. All 18 trials, paired by architecture and seed across resolutions. Lines join individual fitted models; they are not confidence intervals or ensemble results. A and B share an AUROC scale. C shows calibrated test Brier scores (lower is better). Selection and source-test rankings differ, so retrospective choice of the best test result would violate the protocol.

The combined development-only selector chose the 320-pixel EfficientNet-B0 ensemble, whose mean seed selection AUROC was 0.999085, versus 0.998566 for the 224-pixel ResNet18 ensemble. Its temperature was 1.34056 and frozen cutoff 0.83170. Test confusion counts were TN 154, FP 77, FN 1 and TP 386. These aggregate counts match the 224-pixel ensemble, but the two models do not make identical image-level predictions.

Metric	320-pixel ensemble (95% interval)	320 minus 224 (95% paired interval)
AUROC	0.9640 [0.9470, 0.9787]	0.0059 [-0.0055, 0.0170]
Average precision	0.9672 [0.9450, 0.9840]	0.0057 [-0.0079, 0.0208]
Accuracy	87.38% [84.42, 90.17]	0.00 pp [-1.37, 1.31]
Sensitivity	99.74% [99.16, 100.00]	0.00 pp [0.00, 0.00]
Specificity	66.67% [60.87, 72.73]	0.00 pp [-3.57, 3.37]
Brier score	0.1395 [0.1112, 0.1666]	0.0084 [-0.0012, 0.0183]

The paired analysis used 1,000 group-bootstrap replicates on the same 618 images and 426 filename groups. All paired intervals include zero; this neither establishes improvement nor proves equivalence. The zero-width sensitivity-difference interval reflects identical observed decisions among pneumonia-labeled images in this sample, not absence of uncertainty in future sensitivity. No multiplicity correction or retraining was performed. The 320-pixel ensemble had a higher test Brier score than the 224-pixel ensemble (0.13948 versus 0.13105), despite its higher AUROC point estimate.

Within the 320-pixel ensemble, calibration reduced test Brier score from 0.14313 to 0.13948 and log loss from 1.15973 to 0.78876. Calibration therefore behaved differently across arms: it worsened these source-test measures for the 224-pixel ensemble. Raw-score threshold 0.5 gave the 320-pixel ensemble 84.79% accuracy, 100.00% sensitivity and 59.31% specificity. Calibration scores are not established patient-level disease probabilities.

An exploratory development-selection stability analysis was specified after the 224-pixel test result was observed and before the 320-pixel test result. It resampled the 270 selection groups jointly across six

architecture-resolution candidates, averaging their three fixed seed AUROCs. The frozen 320-pixel EfficientNet-B0 choice won 852 of 1,000 replicates; 224-pixel ResNet18 won 112, 320-pixel ResNet18 won 20 and 224-pixel DenseNet121 won 16. Nine replicates had tied maxima resolved by the prespecified ordering. These frequencies describe selection-sample sensitivity; they are not posterior probabilities or evidence of external superiority. Checkpoint selection used this same partition, and no retraining occurred, so selection optimism remains. The chosen model was not changed.

Calibration-only threshold resampling for the 320-pixel ensemble yielded 2.5/25/50/75/97.5 percentiles of 0.60672/0.60672/0.83170/0.84011/0.89223 across 1,000 replicates, with eight distinct thresholds. Temperature was held fixed and the original cutoff retained. This descriptive spread is not a confidence interval for a clinically optimal threshold.

The 320-pixel CPU verification passed in 433.3 seconds (Kaggle version 360906861). All nine checkpoints loaded strictly and passed a forward check. Decoded-pixel identities matched for all 618 test images. Fresh selected-ensemble inference changed 0 classifications; maximum score difference was 0.00000464, below tolerance 0.0001. Artifact validation independently reconstructed the ensemble and all reported trial test metrics. The paired analysis completed in 66.2 seconds (version 360907021). These are engineering checks on the same images, not external validation.

Post hoc frozen-error analysis (Kaggle version 360916945) found 531 images classified correctly by both ensembles and 69 incorrectly by both. Each ensemble alone classified nine additional images correctly. All 18 disagreements were NORMAL-labeled: 145 normal images were correctly classified by both and 68 were false positives for both. Both ensembles detected the same 386 of 387 pneumonia-labeled images. Among source filenames, 239 of 240 bacterial-labeled and all 147 viral-labeled pneumonia images were detected by each model. These are descriptive binary-classifier counts, not evidence of bacterial-versus-viral diagnostic performance or guaranteed future sensitivity. Filename annotations were not independently confirmed. One proxy group spans both pneumonia filename subgroups; subgroup group counts must not be summed as independent patients. No model or cutoff was changed.

Discussion and limitations

The preliminary results illustrate that operating thresholds can change observed accuracy and specificity without improving ranking or calibration. A credible analysis must therefore distinguish performance of the score from performance of a particular decision rule. The new experiment separates model selection from calibration, but its small calibration cohort and conservative grouping introduce uncertainty that additional GPU time cannot remove.

The source test set was previously inspected. Its results are exploratory, the filename groups are proxies, and the study lacks independent clinical adjudication, documented acquisition-site strata and adequate demographic information for a fairness analysis. Exact-pixel duplicate removal cannot rule out transformed duplicates, pretraining overlap or acquisition shortcuts. The binary dataset target does not establish etiology, severity, treatment need or validity for non-chest images.

The similarity screen found 814 candidate cross-partition pairs, including 269 involving test images, among 5,371 retained images. Limited contact-sheet inspection showed visible differences in reviewed pairs and did not confirm image duplication. Most candidates remain unreviewed. Perceptual similarity can reflect shared anatomy, and visual differences do not establish different patients. These counts must not be presented as a measured leakage rate or proof of clean patient separation.

BDCXR-3257 is a candidate external pediatric source [5]. The public record lists CC BY 4.0 and describes contrast enhancement and cropping, which would create preprocessing differences. A documentation-only

inspection of a Kaggle copy [8] confirmed 880 normal, 1,477 bacterial and 900 viral images and 3,257 metadata rows. Filename, Age, Gender and Class fields have no missing values. Included certificates agree with these counts, resolving the numerical discrepancy in the public description; no individual patient identifier is supplied.

The mirror provides no provenance citation and lists its license as unknown. Its included agreement identifies a particular hospital and university research laboratory as parties, limits access to authorized research personnel, and restricts raw-image redistribution. These statements have not been reconciled with the public CC BY listing. Document authenticity and release authority were not independently verified. No external inference was performed, no agreement was accepted, and no claim of applicable ethics approval is made for the present project. External testing remains pending source clarification.

Related pediatric transport work already evaluates BDCXR and VinDr [3]. Differences in development cohorts, preprocessing and model design prevent direct claims of superiority over that study. This report is not a clinical validation or evidence of readiness for deployment. Reporting guidance such as CLAIM 2024 [6] informs transparency; merely referencing the checklist is not certification of completeness.

Reproducibility and publication status

The repository contains training source, Kaggle notebooks, frozen cohort rules, protocol JSON, inference code and artifact validators. Kaggle versions 360875315 and 360888746 are the completed 224- and 320-pixel training runs. Earlier versions 360813730 and 360829588 contain the completed baseline and follow-up. The submitted source hash, runtime versions, split manifest, histories, checkpoints and per-image predictions should accompany the final analysis. Data and weights are excluded from ordinary source control.

The compact-record audit cross-checks trial selections, confusion-derived measures, paired differences and bootstrap winner counts. It does not independently recompute image-level AUROC or uncertainty intervals. Original GPU raw calibration probabilities were not exported. A separate CPU run (Kaggle version 360912398) evaluated all 482 calibration images for each selected ensemble, exported fresh raw scores and independently refit temperature and cutoff. Relative temperature differences were below 0.000002 and absolute cutoff differences below 0.0000014. The selected 320-pixel ensemble passed all checks. The 224-pixel comparator changed one calibration classification, so the combined run correctly failed its strict zero-change criterion after saving both reports.

Boundary inspection (version 360914770) found that this comparator score originally equaled its cutoff exactly; CPU inference moved it lower by 0.000001038. Reading saved scores with round-trip precision reproduced all stored original decisions. The strict calibration failure remains recorded; no cutoff or model was changed. Both models reproduced every test classification. Fresh CPU scores provide additional calibration-fit evidence, but do not recover the missing original GPU raw exports. These checks establish engineering reproducibility within stated numerical limits, not clinical validity.

The work is not submitted to a journal. The source release destination is github.com/bekturovtemirlan000-gif/pneumonia-research-benchmark; source publication does not constitute peer review. Before submission, the responsible human authors must verify all claims and references, establish authorship and affiliations, determine applicable ethics requirements for their setting, disclose AI assistance and funding or conflicts accurately, and assess novelty and venue fit. No institutional approval, informed-consent exemption, author contribution or conflict declaration is invented here.

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